Uddhav P. Gautam

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Seeking full-time roles across software architecture, technical leadership, and applied research

Software veteran and open-source enthusiast — a decade as Software Engineer, Tech Lead, and Architect

I build reliable, modular, scalable, and maintainable software using proven development best practices

Research Interests

Security: supply-chain compromise; fail-closed evidence semantics; BLE and embedded post-compromise defense; mobile forensics; SCADA monitoring.

Systems: sandboxed eBPF extensibility; reboot-free OTA security-policy lifecycles; verifier-to-runtime conformance.

Software Engineering: verifier independence for AI-assisted development; externalized evidence contracts.

Publications

Manuscripts Under Review

- **Uddhav P. Gautam**, et al. "Breaking Self-Referential Trust in Source Releases: A Parity Gate for OS Package Supply Chains." *Ready for submission, USENIX Security 2027.* Artifact: *link to be added*
- **Uddhav P. Gautam**, et al. "Breaking Self-Referential Trust with AIE Verification in AI-Assisted Development and Release Assurance." *Under review, ICSE 2027.* Artifact: figshare
- **Uddhav P. Gautam**, et al. "ZeroDown: Firmware-Decoupled OTA Security-Policy Updates for MCU-Class BLE Peripherals." *Under review, EuroSys 2027.* Artifact: *link to be added*
- *Sahil Dudani*, **Uddhav P. Gautam**, et al. "Ghost in the Device: The Parallel World of Spy Apps Forensics." *Under review, Forensic Science International: Digital Investigation, Elsevier.*
- *Ashraf Saeed*, **Uddhav P. Gautam**, et al. "AI-Code Verification Readiness in CS1: A Seeded-Flaw Diagnostic." *Under review, Koli Calling.*

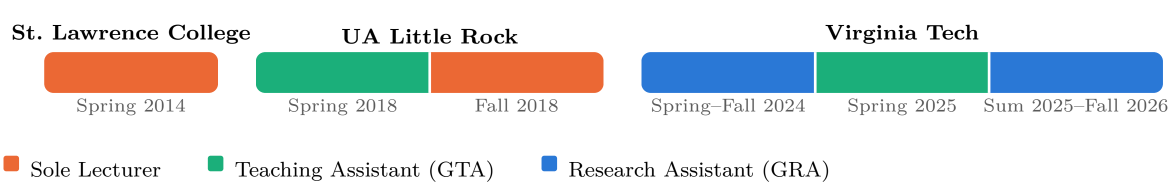
Peer-Reviewed Publications

- Milo Craun, Khizar Hussain, **Uddhav P. Gautam**, et al. "Eliminating eBPF Tracing Overhead on Untraced Processes." *Proc. ACM SIGCOMM Workshop on eBPF & Kernel Ext.*, 2024. DOI: 10.1145/3672197.3673431
- Syed Akailvi, **Uddhav P. Gautam**, et al. "HELOT—Hunting Evil Life in Operational Technology." *IEEE Trans. Smart Grid*, vol. 14, no. 4, pp. 3058–3071, Jul. 2023. DOI: 10.1109/TSG.2022.3222261

Teaching & Research Experience

Courses Taught

- **Sole Lecturer** — Android Apps Development, upper-division (UA Little Rock, Fall 2018); Discrete Mathematics (St. Lawrence College, Nepal, Spring 2014)
- **Graduate Teaching Assistant** — Java Programming (Virginia Tech, Spring 2025); Operating Systems and Microprocessor Programming (UA Little Rock)



Software

Founder and sole developer, RGR Innovate LLC (2024 – Present). Three free, ad-free Android applications published as a public service on Google Play.

- **Earthquakes by RGR**: real-time seismic mapping over USGS GeoJSON feeds with location search and radius/magnitude filtering (Google Maps, Google Places)
- **SmartConvert**: fully offline multi-tool for document scanning with edge detection, image/unit/color conversion, screen ruler, and ARCore room measurement (no accounts, no network required)
- **Passport & ID Photo Studio**: print-ready passport, visa, and ID photos for 155+ countries using on-device face detection, background removal, and automated compliance checks

Sr. Android Developer, Viper Design LLC *Mar 2020 – Feb 2021*

- Built the full IoT software stack for Shark robot-vacuum products, from embedded firmware to cross-platform mobile applications
- Improved maintainability and testability by re-architecting the Android app around modular design, dependency injection, and serialization patterns
- Strengthened production stability through performance tuning, code shrinking, crash analytics, and automated testing

U.S. Army Reserve *Apr 2019 – Apr 2023*

- Served four years and scored in the 96th percentile on the Armed Forces Classification Test

***Prior roles**: 3 years as Android Developer; before that, 4 years as System/Network Administrator after earning Microsoft and Ethical Hacking certifications.*

Education

Ph.D. Computer Engineering, Virginia Tech (GPA: 4.0/4.0) *2024 – Present*
Advisors: Haining Wang, Randy Marchany

M.S. Computer Science, UA Little Rock (GPA: 3.58/4.0) *2015 – 2017*

B.S. Computer Science, Tribhuvan University, Nepal (GPA: 3.91/4.0) *2009 – 2013*

Honors & Awards

- **UA Little Rock** — 1st Place: Complete Societal Award, Societal Impact Award, and Best Graduate Project Award; 2nd Place: Personal Achievement Award and TechLaunch Entrepreneurial Team Projects Competition
- **Tribhuvan University, Nepal** — Top-5 Final-Year Project Award, Nepal Telecom Authority (NTA)

Technical Expertise

Systems & Kernel

- eBPF (TC, XDP, tracing, uBPF sandbox)
- Linux kernel development
- C/C++, Python

Embedded & IoT Security

- BLE protocol security
- Zephyr RTOS, Nordic nRF52840
- Authenticated OTA updates
- Runtime policy enforcement

Security & Assurance

- Software supply-chain security
- Reproducible builds, SBOM, SARIF
- Trivy, OWASP Dep.-Check, MVT
- Mobile forensics (Cellebrite UFED)
- SCADA (DNP3, Modbus, IEC 61850)

Software & Infrastructure

- Kotlin, Java, Android development
- DevSecOps, Docker
- AWS, GCP, Firebase
- Elasticsearch, Logstash, Filebeat

Certifications

Software Architecture, CI/CD & DevOps, Deep Learning (NVIDIA), Java/Kotlin Design Patterns, React Native, Scrum Master, Decision Making, Delegating Tasks, and multiple Microsoft certifications (full list on LinkedIn).

References

- **Haining Wang** Prof., Bradley Dept. of Electrical & Computer Engineering, Virginia Tech, hnw@vt.edu, (757) 813-4390
- **Randy Marchany** CISO & Director, IT Security Lab, Virginia Tech, marchany@vt.edu, (540) 250-7681
- **Brad Tilley** Director, Security Architecture & Red Team, Virginia Tech, btilley@vt.edu, (540) 231-3133